**Last Prompt: Operationalising Partial Perspectives in Decision Intelligence Training**

*Design and Mechanics of a Skin-Modular Foresight Engine*

**Miranda Kelly and Jonathan Kelly**

*Independent Researchers*

*Corresponding author: artikelly@hotmail.co.uk*



*Working Paper — Draft for Review*

May 2026

**Author Note:** The system described in this paper is available in beta at https://last-prompt.com. Practitioner-facing documentation of the architecture, evaluation rubric, and all three skins is available there. The authors' background is in financial services and professional development; the observations in Section 2 draw on direct practitioner experience across both sectors.

**Abstract**

Contemporary organisations have systematically removed the conditions under which consequential decision-making capacity develops. Credentialism implies readiness without conferring the experience it signals. Bounce culture rewards deferral over commitment. KPI architectures narrow professional attention to measured perimeters. Translation gaps prevent clear reasoning from reaching executable form. This paper presents Last Prompt, a domain-agnostic decision intelligence engine that provides calibrated, consequence-bearing reasoning practice under conditions of partial information. The engine separates a fixed evaluation layer — comprising a six-criterion rubric, environmental dimension mechanics, asymmetric feedback, and domain-pressure event selection — from an interchangeable domain layer that defines the narrative world, advisor characters, and stat taxonomy. Three implementations (post-collapse settlement, corporate crisis, civilisational history) demonstrate that the architecture produces structurally identical cognitive demands across fundamentally different professional contexts. The paper positions the system within deliberate practice theory, situated cognition, reflective practice, and metacognitive development, and specifies the falsification conditions its central claims require. This is a design paper: the architectural and theoretical claims are made with confidence; the developmental claims remain appropriately tentative pending empirical validation.

**Keywords:** decision intelligence, deliberate practice, partial information, consequence-bearing reasoning, AI evaluation, skin-modular architecture, management development, situated cognition

**Subject Areas:** Management Education, Decision Sciences, Learning Technologies, Organisational Behaviour

---

## 1. The Engine: What It Does

Last Prompt is a decision intelligence engine built around a single structural commitment: the quality of reasoning can be evaluated independently of the outcome it produces. Every design choice follows from that premise.

The term *decision intelligence* was introduced by Kozyrkov (2019) to describe the discipline of turning information into better action — a framing that synthesises data science, social science, and managerial science around the act of deciding. Last Prompt operates within this framing but addresses a different layer of the problem. Where Kozyrkov's decision intelligence asks how organisations can use data and AI systems to support better decisions, Last Prompt asks what happens when the decision-maker themselves lacks the cognitive capacity to reason well under uncertainty — regardless of how much data or analytical infrastructure surrounds them. The bottleneck the engine addresses is not informational. It is developmental.

The engine runs across multiple narrative contexts — referred to as skins — each presenting the practitioner with a different world, a different protagonist, and a different set of organisational pressures. The Colony skin places the practitioner in the role of Jon Kelly, leader of a post-collapse settlement navigating resource scarcity, social cohesion, and infrastructure failure. The Corporate Reckoning skin positions the practitioner as a senior executive managing cascading crises across a corporation under pressure. Lockwood places the practitioner at crux points spanning all of human civilisational history — from the earliest transformative material discoveries to the present edge of machine intelligence. The narrative context changes across skins. The underlying evaluation architecture does not.

We use *practitioner* throughout this paper to describe the person engaging with the engine, though the term *player* appears where the immediacy of the decision cycle is the relevant quality. The distinction is intentional: Last Prompt is neither a game nor a passive tool, and the vocabulary reflects that.

### 1.1 The Partial Perspective Structure

Each event is delivered through a single advisor — a character whose professional formation shapes what they notice, what they prioritise, and what falls outside their frame. In the Colony skin, this might be a security specialist reporting a perimeter breach, a medical officer raising an outbreak concern, or a logistics coordinator flagging a supply shortfall. Each advisor reports honestly and completely within their domain. None reports beyond it.

This is not a reliability problem. The security specialist is not withholding information about food supplies. Their expertise has simply constructed a horizon, and what lies beyond it is genuinely outside their picture. The practitioner receives one partial perspective per cycle, plus a seed description of the event itself. No advisor aggregates across domains. No single report gives the full picture.

The remaining perspectives are not absent. They are present in the environmental state: five dimensions whose current values reflect the accumulated consequence of every prior decision. The advisor speaks. The stats register silently.

The cognitive demand is therefore not the management of competing voices. It is something harder: the integration of one articulate, credible, domain-specific recommendation against the mute testimony of a system that is already showing the cost of imbalance. The CISO recommends weekend working to address the breach. Employee wellbeing is already low. No one in the room is making the counterargument. The practitioner must make it themselves — or discover, in the chapter debrief, that they consistently failed to.

The practitioner — embodying the protagonist — must formulate a response from this partial information. That response takes the form of a free-text plan comprising a goal, a set of operational actions, a contingency, and a communication strategy. Critically, the protagonist has no access to the engine's evaluation logic. The practitioner submits the plan without knowing how it will be scored.

### 1.2 The Evaluation Architecture

Once submitted, the plan is assessed by a neutral AI evaluator — referred to in the engine's lore as the Mandate — against a structured rubric of six interdependent criteria: variable awareness, resource allocation, risk anticipation, communication clarity, multi-step planning, and temporal symbiosis. Each criterion is scored zero, one, or two. The maximum score is twelve.

The sixth criterion — temporal symbiosis — evaluates whether the practitioner's reasoning demonstrates awareness of how the current decision echoes prior ones: whether the pattern of choices across time is being actively managed, or whether each decision is being treated as if it arrived in isolation. The concept that underlies this criterion is chronosymbiosis — the principle that every decision exists in a living relationship with the decisions that preceded it and the decisions it will make possible or foreclose. A decision that is technically sound in isolation may be strategically destructive if it ignores what earlier decisions have already committed the system to, or if it consumes resources and options that future decisions will need. Chronosymbiosis is not hindsight. It is the active, real-time management of the temporal thread — the practitioner's awareness that they are not solving a problem but extending a sequence, and that the quality of the sequence matters as much as the quality of any individual node within it.

The rubric is calibrated to be genuinely demanding. A score of twelve requires that no criterion has a meaningful gap. Scores above nine are classified as Strong; scores between five and eight as Adequate; scores below five as Poor. The classification band — not the precise score — determines how the five environmental dimensions shift after each decision. It is the accumulated state of those dimensions, not the band itself, that determines what the world looks like next and therefore which events become possible.

### 1.3 The Environmental State and Event Selection

The engine tracks five environmental dimensions simultaneously, each scored on a continuous scale. In the Colony skin these are sustenance, health, security, cohesion, and infrastructure. In the Corporate Reckoning skin: financial solvency, regulatory compliance, operational capacity, employee wellbeing, and reputational integrity. In Lockwood: cognitive, moral, perceptual, collective, and temporal dimensions of the historical moment. The labels differ by skin. The logic is identical across all three.

After each decision, the band determines how the environmental dimensions shift. A Strong band allows dimensions to rise, with only a small negative cap. An Adequate band produces small changes in either direction. A Poor band can only cause dimensions to fall.

The event selection mechanism then reads the current environmental state and identifies the dimension under greatest pressure — the lowest-scoring dial. The severity of that score determines the type of event that can appear: a dimension in crisis produces only the most serious events in that domain; a dimension under strain produces moderate events; a stable dimension produces routine events; a strong dimension opens social or opportunity events. Each event in the library is tagged by primary domain and severity band. The engine matches the weakest dimension to eligible events for the current chapter and severity level.

A separate mechanism governs Black Swan events — low-probability, high-impact occurrences that bypass the normal domain-pressure logic. Their probability of triggering increases as overall environmental instability rises. A practitioner whose environment has been drifting toward aggregate fragility faces a structurally higher probability of unpredictable disruption, regardless of which specific dimension is lowest.

### 1.4 The Asymmetric Feedback Structure

The protagonist experiences the consequence of each decision as narrative outcome: what happened in the world, how people responded, what changed visibly. The protagonist does not see the rubric breakdown, the precise score, or the causal chain between reasoning quality and environmental movement.

The practitioner sees all of it.

This asymmetry is the engine's core pedagogical mechanism. The practitioner watches the rubric in real time, observes exactly which criteria were met and which were not, sees the precise stat movements, and reads the chapter debrief — a synthesis of the cycle's five decisions that identifies not individual errors but consistent patterns of strength and vulnerability across the full decision sequence. The protagonist lives the consequences of reasoning quality without access to the evaluation. The practitioner learns to recognise the difference between strong and weak reasoning while watching those consequences unfold.

At the chapter level, the debrief surfaces what individual scores cannot: the strategic tendency that has been shaping outcomes across multiple decisions. A practitioner who consistently handles immediate crises well but underweights systemic ripple effects will see that pattern named at chapter close — not as a single poor decision, but as a recurring orientation that the environment has been quietly responding to throughout.

At the run level — across all six chapters and thirty decisions — the engine produces a Mandate Intelligence Report: a complete decision analytics package that operates at the practitioner's cognitive signature level rather than the individual decision or chapter level. This includes an average score and strong rate across all decisions, a Command Signature radar chart showing performance across the six rubric criteria, a Decision Arc plotting score volatility across the full run, performance breakdowns by crisis type, and a prose Mandate Intelligence Assessment that names the practitioner's signature strength and critical gap in plain language. The radar chart is, in effect, a visual representation of the practitioner's observer patch — the shape of what they consistently see and what they consistently do not.

The downloadable journal — a complete archive of all thirty decision entries in the protagonist's voice, each accompanied by its rubric evaluation and outcome overview — serves a different function from the analytics. Where the analytics identify the cognitive signature, the journal records the reasoning trace: thirty consecutive windows into how the practitioner actually thinks under pressure, across an unrepeatable path through the decision environment. Its primary value is not immediate post-run reflection but deeper engagement — in coaching, in facilitated cohort debrief, or in the practitioner's own longitudinal review of how their reasoning patterns have developed across multiple runs.

### 1.5 Non-Reproducibility and the Commitment Structure

Each skin is structured around six chapters, each containing five decision cycles — thirty decisions per run. Because event selection is driven by the current environmental state rather than a fixed sequence, two practitioners running the same skin simultaneously will encounter different events in different orders. A practitioner who maintains strong security but allows cohesion to erode will face a fundamentally different sequence than one whose sustenance dial is the persistent vulnerability. The events themselves are the same library. The path through them is unique to each practitioner's decision history.

This non-reproducibility is not incidental. It is the condition that makes each decision genuinely consequential. A practitioner cannot return to a prior decision point, adjust their reasoning, and observe the difference. The environmental state that produced a given event cannot be reconstructed, because the sequence of decisions that shaped it cannot be undone. Each run is, in this sense, a single unrepeatable thread — more analogous to a professional episode than to a training exercise with a reset button.

The practical implication is that the engine offers considerable replayability — no two runs will follow the same path — but not in the sense that most training tools deploy repetition. Repetition here does not mean repeating the same scenario until the correct response is memorised. It means accumulating decision-making experience across genuinely different situations, each shaped by a different history of prior reasoning.

## 2. The Problem the Engine Was Built to Solve

### 2.1 What Credentialism Replaced

For most of human professional history, competence was established through visible consequence. The apprentice who made an error felt it — in the quality of the work, in the response of the master, in the economic reality of a client lost or a structure that failed. The error was not absorbed by a system designed to contain it. It landed. And because it landed, it taught.

This was not a comfortable learning environment. It was, however, an honest one. The feedback was immediate, specific, and impossible to defer. The gap between intent and execution was visible to everyone in the room, including the person who created it. Competence that survived this process was competence that had genuinely been tested.

Credentialism changed the signal without changing the underlying requirement. A credential communicates that its holder has survived a particular kind of structured assessment — examinations, supervised practice, evaluated performance within a defined curriculum. What it cannot communicate, and does not attempt to, is whether that person has encountered a genuinely consequential decision under real uncertainty, been wrong in a way that reached them personally, and developed the specific capacity that only that experience produces.

This is not a criticism of credentialling systems. They solve a real problem: how to communicate transferable competence across contexts where direct observation is impossible. A hospital cannot observe every medical school. A law firm cannot apprentice every prospective associate for years before hiring. Credentials create a scalable signal where no scalable alternative exists.

The problem is not the credential. It is what the credential implies — and what organisations, rationally, infer from it.

When a credential signals readiness, the organisation that hires on the basis of that credential has a reasonable expectation that the person does not need the kind of consequence-bearing exposure that produces real competence. They are, after all, already qualified. The junior lawyer does not need to feel the full weight of a failed case — they are supervised, reviewed, and protected from outcomes that might damage the client, the firm, and the junior lawyer's own developing confidence. The graduate management trainee does not need to own a decision that fails visibly — they are rotated through departments, evaluated on potential, and shielded from the kind of error that would be professionally survivable at twenty-three and professionally damaging at forty.

The protection is well-intentioned. It is also, over time, developmentally costly in ways that do not become visible until much later.

What was lost when credentialism replaced consequence-bearing apprenticeship was not knowledge. The credentialled professional often has more theoretical knowledge than their pre-credential equivalent. What was lost was the accumulated experience of making real calls under real pressure — of being wrong in a specific and attributable way, sitting with it, learning from it, and going again. That is the experience that builds the capacity the credential implies but cannot confer.

> *One of the present authors, at twenty-one, working in the back office of a financial brokerage, missed a payment of 4.2 million dollars. The consequence was immediate, unambiguous, and entirely attributable. There was no committee to diffuse the responsibility, no system designed to catch the error before it landed, no supervisor who absorbed the outcome on his behalf. The error landed. The learning was permanent. It is the kind of mistake you make once.*

When the same authors ask professional development students today whether they can identify a comparable moment — a decision that was genuinely theirs, that went wrong in a way that reached them directly, that taught them something no course or credential could — the answer is frequently that no such moment exists. Not because the students are exceptionally careful or exceptionally competent. Because the structures around them have been designed, rationally and with good intent, to ensure that consequential errors do not reach junior professionals.

The credential implied they were ready. The organisation built structures consistent with that implication. The structures ensured the reps never happened. The cycle is self-reinforcing and largely invisible to everyone inside it.

This tendency has been formalised in recent years through the proliferation of talent identification programmes within large organisations. These initiatives, typically administered through HR functions, aim to identify high-potential individuals early — mapping their capabilities, accelerating their development, and positioning them for senior roles ahead of the conventional timeline. The criteria used to identify talent are instructive: performance consistency, stakeholder management, cultural alignment, and the absence of significant failure. The talent pool, almost by definition, is composed of people who have not made costly visible errors.

The intention is sound. Organisations want to invest development resources in people most likely to return that investment. The effect, however, is to systematically select for individuals who have been most effectively protected from consequence-bearing experience — and to accelerate their progression toward roles where that experience would have been most valuable. The people identified as highest potential are frequently those who have most successfully navigated structures designed to prevent errors from landing. They arrive at senior positions having been selected, in part, for the very quality that the role now most urgently requires them not to have: an unblemished relationship with consequential failure.

### 2.2 The Bounce Culture

The word ”decision” appears frequently in organisational life. The act of deciding — committing to a course of action in a way that is specific, attributable, and irreversible — appears considerably less often than the word suggests.

What fills the gap is a behaviour so embedded in organisational culture that most of its practitioners do not recognise it as a behaviour at all. They experience it as process. As diligence. As appropriate escalation. It has no universally agreed name, but its mechanics are consistent enough to describe precisely: the decision arrives, is acknowledged, is reframed as requiring further input, broader consensus, more data, or a higher authority, and is passed onward. The problem has been handled. Nothing has been decided.

This is bounce culture. And it is not, at its root, a failure of courage or competence. It is a rational response to an asymmetric incentive structure.

In most organisational environments, the consequences of a visible wrong decision fall harder on the individual who made it than the consequences of a decision that was delayed, diffused, or never formally made at all. The manager who commits to a course of action that fails can be identified, reviewed, and held accountable. The manager who convened a committee, gathered additional perspectives, escalated to senior leadership, and waited for consensus cannot. The outcome may be identical — or worse, because the delay compounded the problem — but the attribution is diffuse. No one decided. Therefore no one failed.

The language of bounce culture is indistinguishable, on the surface, from the language of good governance. *I have raised this with the relevant stakeholders. I have passed this to customer service and am awaiting their response. I have flagged this for the next leadership review. I have ensured the appropriate people are informed.* Each sentence describes an action taken. None describes a decision made. The person speaking them has been busy, conscientious, and entirely unaccountable for what happens next.

When the same authors ask professional development students to describe a recent consequential decision they made at work, the responses are revealing. The most common answers are not decisions at all. They are escalations dressed in the vocabulary of action. *I told the customer. I informed my manager. I submitted the report.* The student experiences these as decision points — moments where they exercised judgment and discharged their responsibility. In most cases they are moments where the decision was successfully transferred to someone else, or to no one in particular, without the student recognising the transfer had occurred.

This is not a criticism of the students. It is a description of the environment that shaped them. In organisations where bounce culture is structurally rewarded — where the consequences of visible failure fall on individuals and the consequences of collective non-decision are absorbed by the system — the rational professional learns, usually without being taught explicitly, that commitment is a liability. The skill that gets developed is not decision-making. It is the sophisticated management of the appearance of decision-making while the actual commitment remains perpetually deferred.

The counter-example is instructive precisely because it is now rare. The same author, in the same back office of the same financial brokerage, observed a trade position identified after morning markets — around 11:30. The position needed to be resolved before a 4pm GBP deadline. For three hours the reasonable hope was that the manager — whose desk faced directly across — would do what he usually did and handle it. At 3:30 it became apparent there was no way out. He pushed three blank trading slips across the divide without explanation. When told there was uncertainty about what to put on them, he said they would have a problem. A single clarifying question was eventually asked — whether the position required two futures and one option. He said to write it and see what happened.

The slips were completed. The deadline was met.

What those three hours between 11:30 and 3:30 illustrate is not incompetence. They illustrate bounce culture operating exactly as it does in every organisation that has built structures rewarding deferral over commitment — the entirely rational hope that someone with more authority, more experience, or more ownership of the outcome will absorb the decision before the deadline forces the issue. What the manager understood, and expressed through three blank slips rather than instruction, is that the information required to act was already present. The barrier was not knowledge. It was the willingness to commit.

What that manager would find few organisations today willing to replicate is his method. The liability is too visible, the regulatory environment too demanding, the HR frameworks too carefully constructed to ensure that junior employees are protected from exactly the kind of exposure he was providing. His approach has been designed out. The capacity it produced has not been replaced.

That moment is the direct origin of one of the engine's core structural constraints: the practitioner may ask the advisor one clarifying question before they must decide. Not because additional information is unavailable, but because the discipline of committing with incomplete information — rather than deferring until certainty arrives — is precisely the capacity the engine is designed to develop. Certainty does not arrive before consequential decisions. It arrives, if at all, long after them. The one question rule makes that reality structural rather than optional.

### 2.3 The KPI Tunnel and the Succession Problem

Bounce culture does not exist in isolation. It is sustained and reinforced by a measurement architecture that has become the dominant framework for managing professional performance across most large organisations: the Key Performance Indicator system.

KPIs solve a genuine problem. In organisations of any significant size, direct observation of individual performance becomes impossible. A senior leader cannot watch every person in every team make every decision. A scalable proxy is needed — something that converts complex, contextual performance into a number that can be tracked, compared, and managed at distance. KPIs provide that proxy. They define what good performance looks like in measurable terms, attach those measures to individuals, and create accountability structures around the numbers.

The problem is not the measurement. It is what sustained measurement at the individual level does to the cognitive architecture of the person being measured.

When performance is defined by a set of specific metrics, the rational professional learns — again, usually without being taught explicitly — to attend to what is measured and to treat what is not measured as background. This is not cynicism. It is how human attention works under sustained evaluative pressure. The domain of the KPI becomes the domain of professional concern. What falls outside it falls outside the frame.

This is the KPI tunnel. It is not a personality trait. It is not a generational attitude. It is the predictable cognitive consequence of living inside a measurement system that defines the perimeter of what counts as your problem.

Some organisations have attempted to address the tunnel effect through mandatory internal rotation — moving professionals across functions on a fixed cycle, typically every four to six years. The intention is to build breadth of perspective and prevent the kind of deep siloing that a single-function career produces. The effect, however, is not convergence. It is the sequential accumulation of parallel models.

A professional who has spent five years in operations and five years in finance has not developed the capacity to synthesise operational and financial perspectives simultaneously. They have learned to perform within two different tunnels, one after the other. At no point in that rotation were they required to hold both frames concurrently and produce a response that was coherent across both — because the rotation itself moves them out of one defined perimeter and into another. The measurement system changes. The behaviours it rewards change. The tunnel moves. Its structure does not.

The practitioner who emerges from a rotation programme has visited multiple domains. They have not learned to think across them. That is a different cognitive skill, and sequential exposure to separate measurement architectures does not produce it.

The senior manager who observes that younger colleagues lack ownership, fail to see the bigger picture, or treat adjacent problems as someone else's responsibility is frequently diagnosing the output of their own organisation's measurement design. The younger manager is not disengaged. They are doing exactly what the system trained them to do — performing within the defined perimeter, escalating what falls outside it, and measuring their own success by the metrics that will determine their next review. The environmental awareness that the senior manager takes for granted — the capacity to notice the food stat declining while solving the immediate crisis — was not taught. It was acquired through years of operating in environments where the perimeter was less defined, where errors that crossed domains landed on the person who missed them, and where the boundary between your problem and someone else's was established by consequence rather than job description.

The senior manager did not develop that awareness through instruction. They developed it through exposure — often painful, sometimes career-defining — to the kind of consequence-bearing experience that contemporary organisational structures have largely eliminated from junior professional life. They remember what it felt like when something that was technically outside their remit became catastrophically their problem anyway. The younger manager has never had that experience. Not because they are less capable, but because the structure protected them from it.

This is the succession problem — and it compounds quietly across organisational hierarchies in ways that only become visible at moments of genuine crisis.

As long as the structure holds — as long as KPIs define the perimeter, bounce culture absorbs the decisions that cross it, and talent programmes accelerate the people who have most successfully navigated both — the gap remains invisible. Performance reviews look healthy. Metrics are met. The organisation functions. The people inside it develop sophisticated competence within their defined lanes and genuine inexperience with everything outside them.

The moment the structure fails to hold is the moment the gap becomes visible. A crisis arrives that crosses domains simultaneously — financial, reputational, operational, human. The decision cannot be bounced because there is no higher authority available, or because the higher authority is the one asking for direction. The KPI tunnel offers no guidance because the problem does not sit inside any single metric. The bounce culture has no committee to convene because the deadline is now.

The people in that room have credentials. They have performance histories. They have been identified, in many cases, as high potential. What they have not had is the experience of a 4pm deadline with three blank trading slips and no one coming to absorb the decision for them.

The organisation that built the structures protecting them from that experience now needs them to have it. And the gap — invisible for years, sustained by well-intentioned design — surfaces at exactly the moment it is most costly.

**2.4 The Translation Gap**

The three preceding arguments share a common structure: they describe conditions that prevent a decision from being made. The credential gap means the reps were never accumulated. The bounce culture means the commitment was never owned. The KPI tunnel means the environmental context was never attended to. Each is a failure that occurs before the decision is fully formed.

The translation gap is different. It occurs after.

A decision has been reached. The commitment is genuine. The environmental awareness is present. The person at the desk — or the head of the table, or the terminal — knows what needs to happen. The gap that remains is the distance between the clarity of that knowledge and the precision with which it can be expressed in a form that others can act on.

This gap is almost impossible to see from the inside. The person who has made the decision knows what they mean. Every unstated assumption, every piece of context that did not make it onto the page, every ambiguity in the instruction — the decision-maker fills these in automatically, because they were present for the thinking that produced them. The plan reads clearly to its author. It reads differently to everyone who wasn't in the room when it was formed.

The simplest test is also the most revealing: could someone who just walked in — knowing nothing about the situation, the history, or what was meant — read what was written and know exactly what to do first, without asking what was intended? In most cases the honest answer is no. Not because the decision was wrong. Because the decision was never fully externalised. What exists on the page is the residue of a thought process, not a complete and executable plan.

This matters in isolation. It compounds across organisations.

A slightly vague plan gets executed slightly differently than intended. Slightly different execution produces slightly different conditions. The next decision is made in response to those conditions — conditions that were shaped, in part, by the gap between what was meant and what was communicated. Each cycle the gap has another opportunity to compound. The organisation that appears to be executing a coherent strategy is frequently executing several slightly different interpretations of it simultaneously, none of which is quite what anyone decided.

The translation gap is also the most democratically distributed of the four problems described in this section. The credential gap is most acute in younger professionals. The bounce culture is most visible in mid-level management. The KPI tunnel is most damaging in organisations that have had time to build deep measurement architectures. The translation gap belongs to everyone —

to the senior leader whose strategic vision produces three different interpretations in the room, to the junior manager whose operational plan leaves the team unclear about priorities, to the experienced professional whose decades of contextual knowledge have made certain things so obvious that writing them down no longer feels necessary.

It is also the gap that credentialism is least equipped to address. Credentials assess whether the holder understands the subject. They do not assess whether the holder can convert that understanding into instructions that a stranger could execute without supplementary explanation. The examination is sat alone. The essay is written for an assessor who shares the educational context. The supervised practice is observed by someone who already knows what good looks like. None of these conditions replicate the moment when a plan must be read cold, by someone with no prior context, under time pressure, and acted on immediately.

The result is a professional class that is frequently highly capable of thinking clearly and inconsistently capable of communicating that thinking in executable form. The gap between the two is the last place the credential looks, and therefore the last place the professional thinks to look themselves.

What closes it is the same thing that closes the other three gaps: volume, honest feedback, and conditions complex enough to matter. Writing a plan under real pressure. Being scored not on the outcome, not on the credential, not on the intention — but on whether what was written could be acted on by someone who wasn't in the room when it was formed.

That is what the Mandate was built to provide.

### 2.5 Methodological Positioning

This paper follows a design science research approach as articulated by Hevner, March, Park, and Ram (2004). Design science produces knowledge through the construction and evaluation of artefacts designed to address identified organisational problems. The contribution is the artefact itself — its architecture, the principles that govern its operation, and the theoretical grounding that explains why those principles address the observed problem — rather than an empirical measurement of effect size. The four gaps described in Sections 2.1 through 2.4 constitute the problem space. The engine described in Section 1 and the design principles elaborated in Section 3 constitute the artefact. The theoretical grounding in Section 5 establishes the evaluative framework against which the artefact's design coherence can be assessed.

The paper does not claim empirical validation of developmental outcomes. It claims architectural coherence — that the system's mechanisms are structurally aligned with the conditions that established learning science identifies as necessary for skill development, and that the design choices address the specific failure modes observed in professional practice. The distinction between demonstrated architectural coherence and empirically validated developmental effect is maintained throughout, and the conditions under which the central claims would be falsified are specified explicitly in Section 8.

## 3. Design Principles: Why It Works This Way

The engine described in Section 1 did not arrive at its current architecture through theoretical derivation. It emerged from a specific set of observed failures — the four gaps described in Section 2 — and from the practical constraints that those failures imposed on what an honest evaluation system could look like. Each design principle in this section is a response to a specific failure mode. None of them are stylistic preferences. They are structural necessities.

### 3.1 Professional Horizon: Why Partial Perspectives Are a Feature

The advisors in Last Prompt are not unreliable narrators. They do not withhold information strategically, shade their reporting to serve departmental interests, or frame events to manipulate the practitioner toward a preferred outcome. They are, within their domain, honest and complete.

What they are not is comprehensive. The security specialist who reports a perimeter breach sees the perimeter clearly — the vulnerabilities, the resource requirements, the risk profile, the likely escalation path. What they do not see, and cannot be expected to see, is how addressing the perimeter breach will interact with the colony's sustenance levels, cohesion dynamics, or infrastructure constraints. That is not a failure of character or commitment. It is the predictable consequence of deep expertise. The further a professional develops within a domain, the more precisely they see within it — and the more naturally the adjacent domains recede from active attention.

This is what the engine means by professional horizon. It is not bias in the sense of motivated distortion. It is the cognitive boundary that expertise naturally produces — the edge of the frame that forms around anyone who has spent years developing genuine mastery in a defined area. The CISO thinks in attack surfaces and compliance obligations. The CFO thinks in liquidity and covenant risk. The operations director thinks in throughput and capacity. All three are right about what they see. None of them is seeing the same thing.

The practitioner receives one advisor per event — a single professional perspective, honestly and completely reported within its domain. The remaining perspectives are not absent. They are present in the environmental state: five dimensions whose current values reflect the accumulated consequence of every prior decision. The advisor speaks. The stats register silently.

The cognitive demand is therefore not the management of competing voices. It is something harder: the integration of one articulate, credible, domain-specific recommendation against the mute testimony of a system that is already showing the cost of imbalance. The CISO recommends weekend working to address the server breach. Employee wellbeing is already critically low. No one in the room is making the counterargument. The practitioner must make it themselves — or discover, in the chapter debrief, that they consistently failed to.

The decision-maker's task is not to identify the most credible advisor and follow their recommendation. It is to hold the advisor's partial picture alongside the silent testimony of the environmental state and produce a response that is coherent across both. This is the cognitive demand the engine is designed to train. And it has a direct consequence for what an honest evaluation architecture can look like.

### 3.2 Free Text as Structural Necessity

Because the professional horizon structure means no advisor aggregates across domains, and because the overlap between what the advisor knows and what the practitioner knows is entirely determined by the practitioner's own background, experience, and reasoning — the engine cannot anticipate in advance what a strong response looks like.

This rules out multiple choice as an evaluation format. Not as a stylistic preference. As an architectural impossibility.

Multiple choice requires that the person designing the assessment knows, before the practitioner responds, what the range of plausible answers is — and can construct options that meaningfully distinguish between stronger and weaker reasoning. In a system where the quality of the response depends on how the practitioner synthesises a partial perspective received from a specific advisor, in a specific environmental state, at a specific point in a unique decision history — the space of valid strong responses is too large, too contextual, and too dependent on what the practitioner brought to the situation before they sat down. No option set can honestly contain it.

More precisely: the thing being evaluated is the synthesis itself — a production, not a selection. How the practitioner integrated the advisor's partial perspective with their awareness of the current environmental state, their anticipation of second-order consequences, their communication of priorities to people with different stakes. That synthesis cannot be anticipated in a pre-imagined list. It is something the practitioner generates from the materials available to them, in the time available, under the pressure of irreversible consequence.

Free text is therefore the only format in which that production can be honestly examined. The practitioner must write the plan. Not identify the best plan from options someone else constructed. Not indicate which elements they would prioritise from a checklist. Write it — in language specific enough that someone who just walked in could read it and know exactly what to do first.

This is also where the translation gap from Section 2 becomes measurable. In a multiple choice format, the translation gap is invisible — the practitioner selects an answer and the ambiguity of their actual thinking never surfaces. In a free text format, the gap between what was meant and what was written is present in every submission. The Mandate evaluates not what the practitioner intended but what they produced. The rubric criterion for communication clarity is not asking whether the practitioner had a clear idea. It is asking whether that idea made it onto the page in a form that others could act on.

The professional horizon makes free text necessary. Free text makes the translation gap visible. The Mandate makes it measurable. These three elements are not three separate design decisions. They are one decision, with three components, each of which requires the others.

### 3.3 Asymmetric Feedback: The Metacognitive Loop

When the practitioner submits a plan, two things happen simultaneously. The world responds — the narrative consequence unfolds, the protagonist experiences what changed, what people said, what the settlement or corporation or historical moment looks like after the decision landed. And the Mandate scores the reasoning that produced it.

The protagonist sees only the first. The practitioner sees both.

This asymmetry is not a game mechanic borrowed from entertainment design. It is the structural solution to a specific pedagogical problem: how do you give a practitioner honest feedback on the quality of their reasoning without removing the experience of consequence that makes the reasoning feel real?

Most evaluation environments resolve this tension by choosing one side. Immediate feedback systems — the kind that tell you whether your answer was right before you've had time to feel the weight of being wrong — prioritise learning efficiency over consequential realism. They are educationally useful and experientially hollow. The practitioner learns the correct answer but never develops the capacity to evaluate their own reasoning in real time, because the evaluation always arrives from outside before the internal process has run its course.

Consequence-bearing environments — the trading floor, the boardroom, the crisis room — resolve it the other way. The consequences are real and the feedback is honest, but it arrives slowly, indirectly, and without the precision needed to connect a specific reasoning failure to a specific outcome. The colony doesn't announce that it's running low on food because of the decision made three cycles ago. It just runs low. The practitioner who caused it may never make the connection.

The asymmetric feedback structure holds both simultaneously. The protagonist lives in the consequence-bearing environment — experiencing outcomes without access to the evaluation logic, unable to trace the precise causal chain between reasoning quality and what the world is now doing. The practitioner lives one level above — watching the rubric score arrive, seeing exactly which criteria were met and which were not, observing the stat movements, and reading the precise causal language the Mandate uses to connect the quality of the reasoning to the direction of the world.

The practitioner is therefore doing two things at once. They are making decisions as the protagonist — under pressure, with partial information, against an irreversible clock. And they are learning to evaluate reasoning quality in real time — developing the internal standard that allows a decision-maker to say, before the outcome arrives, whether the reasoning behind a plan is sound. That internal standard is what the credential never tests and the consequence-bearing environment never makes explicit. The asymmetric feedback structure makes it explicit every cycle, without removing the consequence that gives it meaning.

### 3.4 Environmental Stats as Independent Accountability

The rubric evaluates the quality of reasoning about the presented problem. The environmental stats evaluate something the rubric cannot reach: whether the practitioner was attending to the right conditions in the first place.

These are separate cognitive skills. The engine holds them separately and scores them independently — and the independence is the point.

A practitioner can score Strong on every rubric criterion across an entire chapter. Variable awareness demonstrated, resource allocation specific, risk anticipated, communication clear, multi-step planning evident, temporal symbiosis present. Twelve out of twelve, cycle after cycle. And the simulation can still end — not because the reasoning was flawed, but because it was applied in isolation from the surrounding environmental conditions that changed what the correct response actually was.

The CISO's recommendation to address the server risk is logically sound. Addressing it by requiring weekend working is logically sound within the presented problem. It is contextually destructive when employee wellbeing is already critically low — because the environment has made the human cost of that decision greater than the technical risk it resolves. The reasoning didn't fail. The frame was too narrow. The practitioner was thinking correctly about the problem as presented, without registering that the environment had already rewritten the terms on which that problem needed to be solved.

The stat that hits zero is always visible. It sits at the top of the screen throughout every decision cycle. It is not hidden, not obscured, not revealed only in retrospect. The information was present. The practitioner simply was not attending to it — because the presented problem was more immediate, more articulate, and more demanding of attention than a number declining slowly in the peripheral frame.

This is contextually unanchored reasoning — and it is the most organisationally costly reasoning failure that the credential system never surfaces, because credentials test performance on presented problems. They do not test whether the practitioner noticed which conditions were already rewriting the terms of that problem.

The environmental stats make this failure consequential in a way that no rubric criterion can. A rubric criterion for variable awareness can reward a practitioner who names the relevant variables in their plan. It cannot hold a practitioner accountable for noticing that a variable not mentioned in the event description is approaching a critical threshold. That noticing is pre-rubric — it happens, or doesn't happen, before the practitioner begins to write. The stats are the only mechanism in the engine that holds the practitioner accountable for what they attended to before they started reasoning, not just for how well they reasoned once they began.

Together, the rubric and the stats create two accountability layers operating at different levels of cognitive demand. The rubric asks: given the problem you engaged with, how well did you reason about it? The stats ask: were you reading the environment accurately when you chose how to engage with it? A practitioner who answers the first question excellently and the second question poorly will, eventually, find the world ending around them while their individual scores remain strong.

#### *3.4.1 The Chapter Debrief as Convergence*

The two feedback channels — the rubric's immediate precision and the stats' cumulative pressure — produce their most important output not at the level of the individual decision but at the level of the chapter.

The chapter debrief synthesises both. It reads across every decision in the five-cycle chapter, identifies the rubric criteria that were consistently strong and consistently weak, maps those patterns against the stat movements they produced, and names the strategic tendency that has been shaping the environment throughout — not as a series of individual errors but as a recurring orientation that the world has been quietly responding to.

This is feedback at a level that neither the rubric nor the stats can produce alone. A practitioner who consistently scores well on resource allocation but poorly on risk anticipation will see that pattern named — not as a score on a specific decision but as a portrait of how this practitioner thinks under pressure: what they reliably do well, where they consistently leave the environment exposed, and why the world moved in the directions it did across the full chapter.

It is also the closest the engine comes to the feedback that the trading floor manager provided with three blank slips and a 4pm deadline — not instruction, not correction, but a precise reflection of what the practitioner's own reasoning has been producing, returned to them in a form they can act on.

The rubric tells the practitioner how well they reasoned this cycle. The stats show them what their reasoning has been doing to the world. The chapter debrief tells them who they are as a decision-maker — and, implicitly, where the next run should focus.

### 3.5 Domain Pressure Event Selection: The World Attends to What You Don't

Most training environments are designed to be fair. Scenarios are distributed across domains, difficulty is managed to maintain engagement, and the practitioner is rarely confronted with the same weakness twice in immediate succession.

The implicit message is that competence is broad and assessment should reflect that breadth.

The Last Prompt event selection architecture is designed around a different premise entirely. The world is not fair. It does not distribute its pressures evenly across domains or calibrate its demands to the practitioner's current confidence level. It attends, with indifferent precision, to whatever is weakest — and sends its hardest events from exactly that direction.

The mechanism is straightforward but its implications are significant. After each decision cycle the engine reads the current environmental state and identifies the dimension under greatest pressure. Events are then selected from the domain that matches that dimension, filtered for the severity the current score implies. A dimension in crisis produces only the most serious events in that domain. A dimension that is stable produces routine events. The world, in other words, is continuously watching where the practitioner isn't — and responding to neglect with escalating pressure from precisely the direction being neglected.

This is not punitive design. It is an accurate model of how complex environments actually behave. Organisational problems do not arrive randomly distributed across domains. They compound in the areas that are not being attended to. The supply chain failure does not emerge from the part of the operation being managed well. The cultural fracture does not develop in the team whose manager is paying close attention. Pressure finds the unattended corner — not through malice, but through the simple mechanics of neglect compounding over time.

The Black Swan mechanism adds a further layer of precision. As overall environmental instability rises — as multiple dimensions drift toward fragility simultaneously — the probability of an unpredictable high-impact event increases independently of which specific dimension is lowest. This models something that scenario-based training consistently fails to capture: that systemic fragility is not the sum of individual domain weaknesses but a property of the whole environment. A practitioner who has been consistently adequate across all dimensions — never failing badly, never excelling — accumulates a different kind of risk than their individual scores suggest. The Black Swan does not arrive because of any single decision. It arrives because the overall environment has been allowed to drift into a state where unpredictability becomes structurally more likely.

The contrast with adaptive difficulty systems is worth making explicit. Adaptive difficulty adjusts to keep the practitioner comfortable — reducing pressure when performance drops, increasing it when performance is strong. The goal is sustained engagement. The domain pressure system adjusts to expose the practitioner's weakest area — increasing pressure precisely where performance has been insufficient. The goal is not comfort. It is the surfacing of the specific blind spot that comfortable training environments are designed to avoid revealing.

A practitioner who emerges from a full run of Last Prompt having maintained all five dimensions above crisis level has not done so by being generally competent. They have done so by noticing, repeatedly, which dimension was beginning to drift — and attending to it before the event selection mechanism began sending crises from that direction. That noticing is environmental awareness. It cannot be taught by instruction. It can only be developed by operating in an environment that consistently and specifically penalises its absence.

### 3.6 Non-Reproducibility: The Commitment Structure

The final design principle is the one that makes all the others consequential.

Every mechanism described in this section — the professional horizon, the free text evaluation, the asymmetric feedback, the environmental stat layer, the domain pressure selection — produces its intended effect only if the practitioner believes the decision matters. A practitioner who knows they can replay the run, revisit a prior decision point, or reconstruct the conditions that produced a particular event sequence will engage differently than one who knows they cannot. The former is practising. The latter is deciding.

Non-reproducibility is the structural guarantee of the latter condition.

Because event selection is driven by the current environmental state rather than a fixed sequence, two practitioners running the same skin simultaneously will encounter different events in different orders. The events in the library are the same. The path through them is unique to each practitioner's decision history. A run that produced a particular sequence of crises cannot be reconstructed — not because the events are randomised, but because the environmental state that called each event into being was itself the product of a unique sequence of decisions that cannot be identically repeated.

This has a practical implication that distinguishes Last Prompt from scenario-based training in a way that matters for the paper's central claim. Scenario-based training deploys repetition as its primary learning mechanism — the practitioner encounters the same scenario multiple times, refines their response, and converges toward the expected answer. The learning is real but it is the learning of a known problem. The practitioner becomes better at that scenario. They do not necessarily become better at the cognitive demand that scenario was designed to represent.

Last Prompt deploys repetition differently. Each run is a genuinely different path through the same environment — shaped by a different history of prior reasoning, calling different events into being, producing different chapter debriefs. The practitioner who runs the engine multiple times is not practising the same decision repeatedly. They are accumulating decision-making experience across genuinely different situations, each one a unique consequence of how they reasoned in the ones before it.

The irreversibility within a run is equally important. The practitioner cannot return to a prior decision point, adjust their reasoning, and observe the difference.

The environmental state that produced a given event cannot be reconstructed because the sequence of decisions that shaped it cannot be undone. Each run is a single unrepeatable thread — consequential in the way that professional episodes are consequential, not in the way that training exercises with reset buttons are.

Heraclitus observed that you cannot step into the same river twice. The river has moved. The practitioner has changed. The conditions that existed at the moment of the original decision no longer exist. Last Prompt is built on that premise — not as a philosophical position but as a structural constraint. The decision was made. The world moved. The only available response is the next decision, in the environment that the last one produced.

## 4. The Skin-Modular Architecture

The engine described in Sections 1 and 3 was not designed for post-collapse settlements. It was not designed for corporate crisis management or for civilisational history. It was designed for consequential decisions made under partial information — and those exist in every professional context where stakes are real, expertise is domain-specific, and the environment does not wait for certainty before demanding a response.

The skin-modular architecture is the structural expression of that scope. It separates the engine into two distinct layers, each of which can be developed and modified independently of the other. The evaluation layer — the rubric, the band system, the environmental dimension mechanics, the event selection logic, the asymmetric feedback structure, the chapter debrief — is identical across every skin. It does not know, and does not need to know, what world it is operating in. The domain layer — the stat taxonomy, the event library, the advisor characters, the protagonist identity, the narrative voice, the lore — is entirely skin-specific. Replacing the domain layer produces a completely different practitioner experience without altering the evaluation architecture by a single mechanism.

This separation is what makes the modularity claim meaningful rather than cosmetic. It is not that the same design team built three different training environments that happen to share some structural features. It is that the same engine runs across three fundamentally different professional contexts because the evaluation architecture was designed, from the outset, to be domain-agnostic.

### 4.1 Colony: The Reference Implementation

The Colony skin is the engine's reference implementation — the context in which the architecture was first fully realised and from which the clearest documentation of each mechanism exists.

Jon Kelly leads the North Valley Settlement in the aftermath of a cascading systems collapse that stripped away the automated infrastructure of the pre-collapse world. The settlement houses approximately one hundred residents. The five environmental dimensions are sustenance, health, security, cohesion, and infrastructure — each tracking a domain of survival that is both immediately legible and genuinely interdependent.

The advisor roster contains six characters whose professional horizons are not narratively implied but structurally defined. Each advisor carries an encoded decision bias — a set of domain weights that determine what they attend to and what recedes from their frame. Joe Edwards, Head of Security, has a security bias of 1.0 and a health bias of 0.3. Veronica Delany, Community Architect, has a cohesion bias of 1.0 and a security bias of 0.4. Charlie Thomas, Agricultural Lead, has a sustenance bias of 0.9 and a security bias of 0.2. These are not personality traits applied loosely to narrative. They are the mathematical definition of a professional horizon — the precise structural encoding of what each advisor reliably sees and what they reliably do not.

This is what makes the partial perspective claim in Section 3.1 verifiable rather than asserted. The security specialist is not characterised as someone who tends toward security thinking. They are defined as someone for whom security has a weight of 1.0 and health has a weight of 0.3. The honest partiality of their report is not a narrative choice. It is an architectural one.

The event library contains approximately 140 discrete events, each tagged by primary domain and severity band, each capable of appearing only when the environmental state calls for it. The protagonist's journal records the narrative experience of each decision cycle in Jon Kelly's voice — spare, physically grounded, carrying the weight of irreversible consequence without ever referencing the rubric that evaluated the reasoning behind it.

### 4.2 Corporate Reckoning and MegaCorp: Proof of Modularity

The Corporate Reckoning skin replaces every element of the domain layer while leaving the evaluation layer untouched.

The protagonist is a senior executive at MegaCorp, a corporation navigating cascading crises — data breaches, shareholder rebellion, regulatory pressure, operational failure, reputational damage. The five environmental dimensions are financial solvency, regulatory compliance, operational capacity, employee wellbeing, and reputational integrity.

The advisor roster is drawn from the corporate C-suite and senior leadership: Kenny the CISO, whose regulatory compliance bias sits at 1.0 and whose employee wellbeing bias sits at 0.3; Stéphane, Head of Learning and Talent Enablement, whose employee wellbeing bias reaches 0.9 and whose regulatory compliance bias sits at 0.4; Sophia, Head of Market Analysis, whose cash flow bias reaches 0.9 and whose regulatory compliance bias sits at 0.3. The same structural encoding of professional horizon that defines Colony's advisors defines MegaCorp's. Different domain language. Identical architecture.

The practitioner's task is structurally identical to Colony: integrate one advisor's partial perspective against the silent testimony of five environmental dimensions, produce a free-text plan that the Mandate scores against the same six-criterion rubric, watch the band move the dimensions, read the chapter debrief at cycle close. The world is entirely different. The cognitive demand is identical.

The Corporate Reckoning skin also surfaces something that the Colony skin's survival context makes less visible: the tension between advisors whose domain biases pull in structurally opposite directions. Kenny's security-first orientation and Stéphane's people-development orientation are not in conflict because either is wrong. They are in conflict because each is right within their domain, and the practitioner must synthesise them against an environmental state that makes no announcement about which domain currently matters more. That synthesis is the engine's core demand — and it operates identically whether the world is a post-collapse settlement or a corporation under crisis.

### 4.3 Lockwood: The Knowledge-Neutral Civilisational Skin

Lockwood serves as the strongest test of the engine's domain-agnostic claim: if the evaluation architecture can produce meaningful reasoning practice at historical crux points without reducing to either historical trivia or presentist rewriting, it demonstrates transferability across the widest possible epistemic distances. It is the engine's most ambitious implementation — and the one that most directly tests the claim that the evaluation architecture is genuinely domain-agnostic.

The practitioner is the Traveller, a figure who arrives at moments when a discovery, a tool, or a capability crosses a threshold — moments when what happens next is genuinely undetermined and the quality of reasoning about it will shape what becomes possible. The skin spans human civilisational history from its earliest recorded decisions to the present edge of machine intelligence: cave paintings, early legal codes, the lodestone, iron smelting, Viking navigation, the Socratic method, movable type, the telescope, electromagnetism, evolutionary theory, the structure of DNA, ENIAC, the first networked packet transmission, and the emergence of feedback in intelligent systems. Each chapter opens at a defined historical or pre-historical crux point and develops through events that share that chapter's temporal and epistemic texture.

The five environmental dimensions in Lockwood are cognitive, moral, perceptual, collective, and temporal — tracking not the survival of a settlement or the financial health of a corporation, but the epistemological conditions of the moment: how clearly the situation is being perceived, how the moral weight of the decision is being handled, what the cognitive framework for understanding it allows and forecloses, how the collective response is forming, and what the timing of the decision is doing to the thread of what comes next. These are the dimensions of any consequential decision at any point in history. They are, in that sense, the most universal domain taxonomy the engine has yet instantiated.

The central design challenge Lockwood faces is one that Colony and Corporate Reckoning do not: the events are grounded in real historical moments, which means a practitioner with greater historical knowledge might be expected to navigate the skin more successfully than one without — not because they reason better, but because they recognise the scenario. The mirror design principle resolves this.

Each event within a chapter preserves the structural dynamics of its historical or civilisational moment without naming the specific facts that would confer unearned advantage. The Dark Ore Awakens describes a forge pushed past bronze temperatures, a material that came out wrong and ugly and brittle, a smith holding something whose weight does not match what his eyes are telling him — without naming iron. The Lodestone Turns describes a stone that always settles in the same direction, a man who works with his hands and cannot speak directly to the people who would understand what he is seeing, a sacred object that is also a direction-finding instrument — without naming magnetism or the compass. Induction Flickers describes a current that appears only at the moment of change, never in the steady state, a single line in a private notebook that implies continuous electrical generation — without naming Faraday or electromagnetic induction.

A practitioner who knows what iron is good for has no advantage over one who does not, because the evaluation is not of historical recall. It is of reasoning quality: how the practitioner integrates the partial perspective they received with the environmental state they are responsible for managing, how clearly they articulate the decision at the crux, how honestly they account for the second-order consequences of the choice available to them. A practitioner's historical knowledge therefore confers no scoring advantage; the Mandate evaluates only the structure and environmental integration of the reasoning applied to the partial information and state presented. The mirror strips the knowledge advantage while preserving the epistemic texture — the weight of the moment, the partiality of the available information, the irreversibility of what happens next.

The skin does not claim to model what actually happened in history, nor does it assert that better reasoning in the simulation would have produced different historical outcomes. It models the cognitive conditions that existed at moments of genuine openness — partial information, competing horizons, irreversible consequence — and trains the practitioner in reasoning under those conditions. Success in Lockwood is therefore measured by the same rubric criteria (variable awareness, chronosymbiosis, etc.) as in Colony or Corporate Reckoning, not by fidelity to historical fact.

The Black Swan events in Lockwood include a set tagged as universal — capable of appearing in any chapter regardless of historical period. The physician marking a fever map, the cartographer receiving instructions to redraw borders, the librarian watching an archive burn, the editor watching a false story spread faster than the correction. These events recur across human history because the dynamics they represent — disease, territorial redefinition, the destruction of knowledge, the acceleration of misinformation — are not historically located. They are structurally permanent. Their appearance in any chapter is therefore not anachronistic. It is accurate.

The advisor archetypes in Lockwood — the Spark, the Wanderer, the Sovereign, the Weaver, the Questioner, the Survivor — are built around distinct cognitive and epistemic orientations rather than professional domains. The Spark attends to what is newly possible. The Sovereign attends to what is at stake for existing authority. The Questioner attends to what is not yet known. Each represents a fundamentally different way of constructing a partial picture of what is happening and what matters. The professional horizon in Lockwood is not the boundary of domain expertise. It is the boundary of an epistemic stance — a particular way of seeing that is as partial, and as honestly held, as any domain specialist's view.

The Mandate scores the Traveller's reasoning against the same six-criterion rubric it uses in Colony and Corporate Reckoning. Whether the decision concerns the moment iron was first recognised as something beyond a failed bronze, or the moment the first networked packet transmission revealed a design assumption that would define the architecture of global communication, the evaluation asks the same questions: were the relevant variables acknowledged, were resources and actors specified, were second-order consequences anticipated, was the communication clear, was the temporal dimension managed, did the reasoning demonstrate awareness of the thread connecting this decision to the ones before and after it. The evaluation architecture does not change. The world it is applied to spans all of human history.

### 4.4 Encoded Horizon, Universal Evaluation

The three skins share a structural property that is worth making explicit before considering what the architecture enables beyond its current implementations.

In every skin, the partial perspective is not a narrative device. It is an encoded structural feature. The advisor's domain bias weights define what they see. The environmental dimension taxonomy defines what the world tracks. The rubric criteria define what counts as strong reasoning regardless of domain. These three elements — horizon encoding, dimension taxonomy, evaluation rubric — are the modular interfaces between the domain layer and the evaluation layer. Change the first two and the world changes completely. The third remains constant.

This is what makes the chapter debriefs from three different skins so revealing when read alongside each other. Colony identifying a persistent security vulnerability despite strong performance elsewhere. Corporate Reckoning identifying the collapse of employee wellbeing under crisis containment pressure. Lockwood identifying a tendency to undervalue immediate operational stability while planning brilliantly for the long term. Three different worlds. Three different domain languages. Three completely different strategic blind spots. One evaluation architecture producing all three diagnoses from the same underlying logic.

### 4.5 The Open Architecture

The separation of evaluation layer from domain layer has an implication that extends beyond the three skins currently in operation.

Any professional context in which the following conditions hold can be instantiated as a skin: decisions must be made under genuine uncertainty; information arrives through partial perspectives shaped by domain expertise or epistemic orientation; the environment has multiple interdependent dimensions whose state is affected by the quality of decisions over time; and the consequence of reasoning quality is felt in the world rather than absorbed by the structure around it.

Those conditions describe every consequential decision environment that exists.

A humanitarian field coordinator managing disaster response balances medical resource allocation against security risk, community cohesion against supply chain constraints, immediate relief needs against donor reporting obligations and long-term programme sustainability. The five dimensions, the advisor structure, the domain pressure event selection, the translation gap — all of it maps directly. The skin changes. The evaluation logic does not.

A crisis negotiator operates with partial information from intelligence analysts, psychologists, tactical advisors, and community liaisons — each reporting honestly from within their professional horizon, none aggregating across all of them. The practitioner must synthesise those partial perspectives into a response that is coherent across all the dimensions the negotiation is simultaneously affecting.

A public health official managing an outbreak response receives modelling data from epidemiologists, capacity assessments from hospital administrators, compliance projections from behavioural scientists, and political risk assessments from communications advisors. Each perspective is honest and partial. The environmental dimensions — infection rate, healthcare capacity, public compliance, economic impact, institutional trust — are interdependent and moving simultaneously.

In each case the domain layer requires significant development — the event library, the advisor characters, the protagonist identity, the narrative voice, the lore that makes the world feel consequential rather than abstract. That development is non-trivial. What it does not require is any modification to the engine that evaluates the reasoning. The Mandate does not need to learn a new domain. It needs only a new world to apply its existing logic to.

This is the contribution that the skin-modular architecture makes beyond the three existing implementations. Last Prompt is not a Colony simulator with two variants. It is a domain-agnostic evaluation architecture for consequence-bearing reasoning practice — one that can be instantiated in any professional context where decisions matter, expertise is partial, and the environment does not wait.

## 5. Theoretical Grounding

The engine described in the preceding sections was not derived from theory. It emerged from a specific observation — that organisations had systematically removed the conditions under which consequential decision-making capacity develops — and from the practical constraints that observation imposed on what an honest training architecture could look like. The theoretical frameworks presented here were not the origin of the design. They are its confirmation.

This sequence matters. A system built to fit a theoretical framework risks becoming an illustration of the framework rather than a genuine response to the problem. Last Prompt was built to address something observed repeatedly in professional practice across multiple industries and national contexts. That the resulting architecture maps precisely onto established bodies of work in learning science, cognitive theory, and theoretical physics is evidence that those bodies of work are describing something real — and that the system built to address it is structurally coherent rather than merely intuitive.

### 5.1 Deliberate Practice and the Conditions for Skill Development

The most influential account of how genuine expertise develops is Ericsson, Krampe, and Tesch-Römer's framework of deliberate practice, established in their foundational 1993 study of expert performance across multiple domains (Ericsson, Krampe & Tesch-Römer, 1993). Their central finding — that expert performance is not primarily a function of innate talent but of accumulated practice under specific conditions — has been replicated across fields as diverse as chess, music, surgery, and athletic performance.

The conditions Ericsson identifies are precise: the practice must be at the edge of current competence, not comfortably within it; it must generate immediate, specific feedback; and it must be repeated with sufficient volume to produce genuine internalisation of the skill. These conditions are demanding to replicate for most cognitive skills, and almost impossible to replicate for the specific skill of consequential decision-making under partial information — because most environments either remove the consequence to make practice safe, or remove the feedback to make the consequence real.

Last Prompt is designed as a deliberate practice environment for exactly this skill. The rubric provides the immediate, specific feedback that Ericsson identifies as essential — not after the run is complete, but after each individual decision, with sufficient precision to connect a specific reasoning failure to a specific criterion. The compounding consequence structure ensures that practice occurs at the edge of competence: a practitioner who is managing the environment comfortably will find the domain pressure mechanism escalating complexity in the area they are least attending to. Thirty decisions across six chapters, each producing a unique path through the event library, ensures that volume does not collapse into repetition of known scenarios — each run accumulates genuine decision-making experience rather than refinement of a memorised response.

The credential gap identified in Section 2 is, in Ericsson's terms, a deliberate practice deficit. The credentialled professional has theoretical knowledge but insufficient reps under the specific conditions that produce expertise. The bounce culture and KPI tunnel described in Section 2 are the organisational structures that ensure those reps remain unavailable throughout a career. Last Prompt provides the reps — privately, repeatedly, and in conditions complex enough to matter.

### 5.2 Situated Cognition and the Irreducibility of Context

Lave and Wenger's account of situated learning establishes that cognition is not a portable, context-independent capacity that can be developed in one setting and transferred intact to another (Lave & Wenger, 1991). Learning is inseparable from the activity, context, and culture in which it occurs. What a practitioner learns in a decontextualised training environment is, at best, knowledge about a skill — not the skill itself as it operates under real conditions.

This has a direct implication for decision-making training. A practitioner who learns to identify the correct answer on a multiple choice scenario has learned something about decision-making. They have not developed the capacity to make decisions under the conditions that make decision-making difficult: partial information, environmental pressure, irreversible consequence, and the specific cognitive demand of synthesising partial perspectives that do not aggregate themselves.

The engine's architecture is a situated cognition design. Every element that might be removed to make practice more comfortable has been deliberately retained because its presence is constitutive of the skill being developed. The partial perspective structure is retained because synthesising partial perspectives under pressure is the skill. The environmental stat layer is retained because attending to conditions that don't announce themselves is the skill. The irreversibility is retained because committing without the option to revise is the skill. Remove any of these elements and the practice environment no longer develops the capacity it is designed to train — it develops a related but weaker approximation of it.

The KPI tunnel identified in Section 2 is, in Lave and Wenger's terms, a situated cognition problem. The tunnel is not a reasoning failure. It is the predictable consequence of developing professional competence in a context — the KPI-defined perimeter — that systematically excludes the environmental awareness dimension of the skill. The capacity cannot transfer from a context that never included it.

### 5.3 Complex Learning and Whole-Task Practice

Van Merriënboer and Kirschner's Four-Component Instructional Design (4C/ID) model addresses a problem adjacent to the one Last Prompt was built to solve: how to train complex cognitive skills that require the integration of multiple knowledge domains, coordination of qualitatively different sub-skills, and transfer to novel situations (van Merriënboer & Kirschner, 2018). Their central insight — that complex skills cannot be decomposed into isolated sub-skills and taught separately without losing the integration that defines the skill — is directly relevant to the engine's architecture. The KPI tunnel described in Section 2 is, in 4C/ID terms, the result of an organisational training design that teaches sub-skills (domain-specific performance within measured perimeters) without ever requiring their integration.

Where Last Prompt departs from the 4C/ID model is in the role of scaffolding. Van Merriënboer's framework prescribes a progression from high support to low support — learners begin with worked examples and procedural guidance that is gradually withdrawn as competence develops. Last Prompt provides no scaffolding at all. The practitioner receives one partial perspective, an environmental state, and a blank text field. The rubric scores the output. The chapter debrief names the pattern. At no point does the system simplify the task, model a correct response, or reduce the cognitive load to make practice more manageable. This is a deliberate design choice rather than an oversight: the capacity the engine trains — reasoning under genuine uncertainty with no model answer available — is precisely the capacity that scaffolding would prevent from developing. The 4C/ID model trains complex skills through supported whole-task practice. Last Prompt trains a specific complex skill — consequential reasoning under partial information — through unsupported whole-task practice, on the grounds that the absence of support is constitutive of the skill being developed.

### 5.4 Reflection and the Asymmetric Feedback Structure

Schön's account of professional knowledge distinguishes between two modes of cognitive engagement with a problem: knowing-in-action, the tacit, real-time competence that practitioners exercise while doing, and reflection-on-action, the deliberate examination of that knowing after the fact (Schön, 1983). His argument is that professional expertise develops through the iterative relationship between these two modes — acting, reflecting on the action, adjusting the knowing, acting again.

The asymmetric feedback structure in Last Prompt instantiates this relationship at the level of each decision cycle. The protagonist operates in knowing-in-action — making decisions under pressure, in real time, with partial information, without access to the evaluation logic. The practitioner operates one level above — able to observe the knowing-in-action as it unfolds and to engage in reflection-on-action immediately and precisely, through the rubric breakdown and stat movements that arrive after each decision.

What the engine adds to Schön's model is that the reflection is not left to the practitioner's own interpretive capacity. It is structured and calibrated. The rubric provides the framework for reflection. The chapter debrief synthesises individual reflections into a pattern-level analysis. The practitioner does not need to construct the reflection from scratch — the engine provides the scaffold within which reflection occurs, and the practitioner's role is to engage with that scaffold honestly rather than defensively.

This is also the theoretical grounding for the one clarifying question rule described in Section 2. Schön's reflective practitioner does not defer action until reflection is complete. They act, observe, reflect, and act again — in rapid iteration. The one question rule prevents the deferral of action in favour of continued information-gathering, which is the cognitive equivalent of bouncing. It forces the practitioner into the action-reflection cycle that Schön identifies as the site of genuine professional learning.

### 5.5 Metacognitive Development

Flavell defined metacognition as cognition about cognition — the practitioner's capacity to monitor, evaluate, and regulate their own thinking processes (Flavell, 1979). The development of metacognitive capacity is widely recognised as a precondition for expert performance in complex domains: the expert not only performs better than the novice but can explain why, identify where their reasoning is strong and where it is vulnerable, and adjust their approach before the outcome makes the adjustment necessary.

The asymmetric feedback structure is, in Flavell's terms, a metacognitive development mechanism. The practitioner who watches their rubric score arrive — who sees exactly which criterion was met and which was not, in real time, before the narrative consequence has fully unfolded — is developing the internal standard that metacognition requires. They are learning to evaluate their own reasoning as it happens rather than retrospectively, which is the precondition for the self-correction that expert performance depends on.

The chapter debrief extends this to the pattern level. Flavell distinguishes between metacognitive knowledge — knowing that one tends to underestimate second-order consequences — and metacognitive regulation — actually catching that tendency in the moment and adjusting for it. The debrief provides the knowledge. Repeated runs of the engine, with the debrief's pattern diagnosis in mind, provide the conditions for regulation to develop. The practitioner who has been told three times that they consistently underweight risk anticipation has the metacognitive knowledge required to attend to that criterion more carefully in the next decision cycle.

Kahneman's distinction between fast, automatic System 1 thinking and slow, deliberate System 2 thinking is also relevant here (Kahneman, 2011). The tunnel reasoning described in Section 2 is a System 1 failure — the presented problem captures automatic attention and the environmental context recedes. The engine's design forces System 2 engagement: the free text format requires deliberate construction rather than automatic selection, the stat display requires active monitoring rather than passive response, and the chapter debrief requires reflective synthesis rather than immediate reaction. The practitioner who runs the engine repeatedly is developing the habit of System 2 engagement in conditions that default to System 1 — which is precisely the condition under which consequential organisational decisions are made.

### 5.6 Complex Systems and Consequential Uncertainty

Snowden and Boone's Cynefin framework distinguishes between complicated problems — those with knowable correct answers that expert analysis can identify — and complex problems — those where the relationship between cause and effect can only be understood in retrospect, and where multiple competing interpretations are simultaneously plausible (Snowden & Boone, 2007). Most training environments are designed for complicated problems. Most consequential decisions occur in complex ones.

The engine is explicitly designed for the complex domain. The event selection mechanism models the indifferent, non-linear pressure that complex environments generate — pressure that does not distribute itself evenly, does not respond proportionally to effort, and does not announce itself before it compounds. The Black Swan mechanism models the emergent, unpredictable disruption that Snowden and Boone identify as characteristic of complex systems operating under sustained instability. The non-reproducibility ensures that the practitioner cannot develop a complicated-domain response to a complex-domain challenge — there is no pattern to memorise, no sequence to optimise.

The credential gap, the bounce culture, and the KPI tunnel described in Section 2 are all, in Cynefin terms, complicated-domain responses to complex-domain conditions. Credentials categorise competence. Escalation seeks the expert with the correct answer. KPIs define the perimeter of the analysable. None of these are pathological in complicated domains. All of them fail in complex ones — where the correct answer does not pre-exist the decision, the expert's horizon is partial, and the perimeter is precisely what the environment is about to cross.

### 6. A Discovered Adjacency: Observer Patch Holography

The theoretical connections described in Section 5 were identified after the engine's architecture was established — confirmations of a structure that emerged from practice rather than derivations from theory. The connection described in this section arrived differently, and that difference is worth stating plainly. It is presented separately from the theoretical grounding because it draws on a framework that operates outside established learning science and decision theory — in theoretical physics — and because the nature of the correspondence is speculative rather than foundational.

Bernhard Mueller's Observer Patch Holography framework proposes that there is no single objective view from nowhere (Mueller, open access). Reality, in OPH, emerges from many limited observers — each with their own finite patch of knowledge — who must remain consistent wherever their patches overlap. No pre-existing god's-eye universe. No single script. Just overlapping perspectives synchronising into the stable structures we observe. Where two observer patches overlap, their descriptions must agree — and it is that enforced consistency, rather than any external ground truth, that produces the stable reality we inhabit.

The Last Prompt architecture was not designed with OPH in mind. It was built from practitioner observation, refined through repeated implementation, and arrived at its current form before Mueller's framework was encountered. When the framework was read, the correspondence was immediate and required no reinterpretation of either system to sustain. Mueller subsequently confirmed on reviewing the mapping that the correspondence was precise rather than approximate.

The structural parallels are as follows. The practitioner in Last Prompt occupies the position of the finite observer — arriving at each decision cycle with no god's-eye view, receiving one advisor's partial patch, and responsible for synthesising coherently across the overlap between that patch and the environmental state. The advisor archetypes are distinct limited patches with different domain weightings and epistemic orientations — each honest within its frame, none comprehensive across all frames. The Mandate's rubric criteria — and particularly the criterion for variable awareness and the criterion for temporal symbiosis — are consistency enforcement mechanisms: they score how well the practitioner's reasoning holds together across the overlapping patches they are integrating.

The Command Signature radar chart produced at the end of each run is, in OPH terms, a visual representation of the practitioner's observer patch — the precise shape of what they consistently see across the six evaluation criteria and, by implication, what consistently recedes from their frame. No two practitioners produce the same radar. Each shape is the signature of a particular way of constructing partial knowledge under pressure.

The journal system in Last Prompt — and most precisely in the Lockwood skin — maps onto OPH's holographic screen concept in a way that the authors find genuinely striking. In OPH, the holographic screen is not the real substrate. It is the record of observer interactions from which the three-dimensional reality is reconstructed — more fundamental than the space it encodes. In Lockwood, the journal is not a summary of what happened. It is the record of how the practitioner reasoned: the plan in free text, the rubric evaluation of its internal consistency, the narrative consequence, and in the Multiverse Weave feature, alternate timeline projections derived from different quality bands of reasoning. The reality of the Lockwood timeline is reconstructed from the reasoning trace. Not the other way around.

This mapping is offered as a discovered adjacency rather than a theoretical foundation. OPH was not the origin of Last Prompt's design. Its precision as a description of what that design is doing suggests that both are pointing at something structurally true about how partial knowledge produces coherent outcomes — whether in physics or in the practice of consequential decision-making under uncertainty. The authors hold this connection with genuine interest and appropriate tentativeness, and welcome engagement from those working directly in the OPH framework.

### 7. What This Is Not

A system that is genuinely novel is at risk of being understood through the categories that already exist. Last Prompt has been consistently misclassified in three directions, each of which is understandable and each of which is wrong in a specific way. Naming them directly is not defensiveness. It is precision.

### 7.1 Not a Game

The surface features of Last Prompt — a protagonist, a narrative world, events that unfold in response to decisions, an arc from opening to conclusion — are features it shares with games. The classification is understandable. It is also the classification that does the most damage to accurate understanding of what the system is for and how it works.

The serious games literature has made significant progress in taxonomising the space where game mechanics serve non-entertainment purposes. Abt's foundational work (1970) established the category. Michael and Chen (2006) mapped the landscape across military, corporate, healthcare, and educational applications. Djaouti, Alvarez, and Jessel's G/P/S model (2011) provides the most systematic classification, sorting serious games along three axes: their gameplay mechanics, their serious purpose, and the market they serve. Last Prompt resists even this more granular taxonomy — not because the taxonomy is wrong, but because its core assumption is that the artefact being classified is fundamentally a game with a serious purpose attached. Last Prompt is not a game with a serious purpose. It is an evaluation architecture with narrative elements that create the conditions under which the evaluation becomes consequential. The distinction is structural rather than terminological.

Games — including serious games — are designed around engagement. Difficulty is managed to sustain motivation. Failure states are designed to be recoverable. The consequence of a wrong decision is typically reversible — the player reloads, adjusts, and tries again. The implicit contract between a game and its player is that the experience should be enjoyable enough to continue.

None of these properties hold in Last Prompt. Difficulty is not managed for engagement — it is calibrated to expose the practitioner's weakest domain. Failure states are not recoverable — the environmental state that produced a crisis cannot be reconstructed, and the decision that produced it cannot be revised. The implicit contract is not enjoyment but honesty: the system will show the practitioner exactly where their reasoning is insufficient, without softening that information to protect their confidence.

The chapter debrief does not congratulate. It diagnoses. A practitioner who finishes a chapter having scored consistently in the Adequate band will read a precise account of the reasoning pattern that produced those scores and the environmental consequences that followed. That account is not designed to make them feel good about their performance. It is designed to make their blind spot visible.

This is not a hostile environment. It is a demanding one. The distinction matters — a hostile environment discourages engagement, while a demanding one develops capacity. But the demand is real, and it is incompatible with the entertainment contract that the word game implies.

### 7.2 Not a Simulation

Simulation implies that the system models reality — that the colony, the corporation, or the historical moment behaves as the real thing would behave, and that success in the simulation is evidence of competence in the corresponding real environment.

Last Prompt does not make this claim and should not be evaluated against it. The Colony skin is not a model of post-collapse settlement management. The Corporate Reckoning skin is not a model of corporate crisis response. Lockwood is not a model of historical causation. None of them are designed to be accurate representations of the domains they inhabit.

What they are designed to be is *epistemically accurate* — accurate to the cognitive conditions of consequential decision-making rather than to the content of any specific domain. The partial perspective structure is accurate to how information reaches decision-makers in real organisations. The environmental stat system is accurate to the way complex systems compound neglect over time. The non-reproducibility is accurate to the irreversibility of real professional episodes. These accuracies are cognitive and structural rather than domain-specific.

The distinction matters for how the system should be evaluated. A simulation is assessed by how closely its outputs match real-world outcomes. Last Prompt should be assessed by how accurately it replicates the cognitive conditions under which consequential decisions are made — and by whether repeated practice under those conditions develops the reasoning capacity it is designed to develop. Those are different evaluation criteria, and applying simulation criteria to a system designed for cognitive accuracy will produce a category error in both directions.

### 7.3 Not a Training Course

A training course has a curriculum — a defined body of knowledge or skill that is transmitted from instructor to student through structured content. The student who completes the course has been exposed to the curriculum. Assessment measures whether the exposure produced the intended learning. The course can be completed.

Last Prompt has no curriculum. There is no body of knowledge to transmit, no correct answers to be learned, no instructor whose expertise is transferred to the student. The Mandate does not teach. It evaluates. The distinction is precise: a teacher tells you what good reasoning looks like; the Mandate shows you what your reasoning produced and scores it against a rubric that defines quality without prescribing content.

The system cannot be completed in the sense that a course can be completed. A practitioner who finishes all six chapters of the Colony skin has not learned everything the engine has to teach — they have generated one unrepeatable thread through a decision environment that would produce a different thread on every subsequent run. The appropriate unit of engagement is not the course but the practice session — more analogous to a training run or a sparring session than to a module or a qualification.

This also means the system has no graduation point. There is no certificate, no badge, no credential that the engine confers. What it confers is the accumulated experience of having made consequential decisions under partial information, been scored honestly on the quality of the reasoning, and seen the environmental consequences of that reasoning play out in a world that did not absorb them. Whether that experience has developed genuine capacity is a question that only future consequential decisions — real ones, in real environments — can answer.

### 7.4 The Correct Frame

Last Prompt is a consequence-bearing reasoning environment with calibrated evaluation. Each word in that description carries weight.

*Consequence-bearing* distinguishes it from evaluation environments where performance has no effect on what happens next. The environmental state moves. The world changes. The next event is determined by what the last decision did to the dimensions the practitioner is responsible for. The consequences are not symbolic.

*Reasoning environment* distinguishes it from content delivery. The system is not transmitting knowledge. It is creating conditions under which a specific cognitive capacity — the synthesis of partial perspectives into coherent, executable, environmentally-aware decisions — can be practised with honest feedback.

*Calibrated evaluation* distinguishes it from both unstructured reflection and automated scoring of predetermined answers. The Mandate applies a structured rubric to free text, scores against criteria that are defined but not prescriptive, and provides feedback that is specific enough to be actionable without being directive enough to substitute for the practitioner's own reasoning.

No existing category contains all three of these properties simultaneously. That is not a marketing claim. It is the explanation for why the system has been consistently misclassified — and why the misclassification matters for anyone trying to understand what it does.

## 8. Open Questions and Limitations

The preceding sections describe a system that is theoretically grounded, architecturally coherent, and currently in beta across three skin implementations. What they do not describe — and what intellectual honesty requires naming explicitly — is the gap between the system as designed and the evidence that it works at the level the paper claims.

### 8.1 The AI Evaluator and Its Limits

The Mandate is an AI rubric evaluator. Its assessments are probabilistic rather than deterministic. Two practitioners who submit plans of equivalent quality may receive different scores depending on how the language model reads the specific phrasing of each submission. Two practitioners who submit plans of different quality may receive similar scores if the weaker plan deploys the vocabulary of strong reasoning without the substance.

This is a real limitation. The rubric criteria are defined with sufficient precision to anchor the evaluation — the calibration standard explicitly states that a score of two requires no meaningful gap, and that borderline cases score one rather than two. But the model applying those criteria is not infallible, and the gap between what the rubric intends to measure and what the model actually measures in edge cases has not yet been systematically audited.

The redraft mechanism — which allows a practitioner who has scored below nine to revise and resubmit with the previous score as a calibration anchor — partially addresses this by giving the model a consistency constraint across two evaluations of the same decision. But it does not eliminate the underlying variability. Systematic rubric calibration across a larger practitioner dataset is a necessary next step before the evaluation claims of this paper can be made with full confidence.

### 8.2 The Translation Gap Measurement Problem

The paper claims that free text evaluation makes the translation gap visible and measurable. This is true in the sense that a plan whose instructions cannot be acted on by a stranger will score poorly on the communication clarity criterion. It is less true in the sense that the Mandate's assessment of communicative precision is itself a language model judgment — one that may be more forgiving of vague language than a real-world colleague who needs to act on the plan under time pressure would be.

The translation gap is not fully closed by the free text format. It is made more visible than it would be in a multiple choice format. Whether the Mandate's assessment of it is sufficiently accurate to produce genuine improvement in real-world communicative precision is an empirical question that beta testing has not yet answered.

### 8.3 Validity Beyond Beta

The claims in this paper about what Last Prompt develops in practitioners are theoretically grounded in deliberate practice theory, situated cognition, metacognitive development, and reflective practice. They are not yet empirically validated at scale.

Beta testing has produced qualitative evidence consistent with the theoretical claims — practitioners who have run multiple sessions report increased awareness of their reasoning patterns, greater attention to environmental dimensions during decision cycles, and more specific plan construction over time. The chapter debriefs consistently surface patterns that practitioners recognise as accurate descriptions of their reasoning tendencies. These are promising signals.

They are not controlled evidence. A properly designed validation study would require a comparison group, pre and post measurement of the specific cognitive capacities the engine claims to develop, sufficient sample size to distinguish signal from noise, and sufficient follow-through to assess whether improvements in engine performance correspond to improvements in real-world decision-making. None of these conditions currently exist.

The paper makes theoretical and architectural claims with confidence. It makes developmental claims with appropriate tentativeness. The distinction is intentional.

### 8.4 The Journal as Facilitation Tool

The downloadable journal — a complete archive of all thirty decision entries in the protagonist's voice, each accompanied by its rubric evaluation and outcome overview — is currently used primarily as a personal record. Its potential as a facilitation tool has not yet been systematically explored.

A practitioner's journal from a completed run contains thirty consecutive windows into how they actually think under pressure, across an unrepeatable path through a decision environment. The pattern visible in a completed journal — the recurring gap between strategic vision and operational specificity, for instance, that appears across iron smelting, the Cuban Missile Crisis, the Encyclopédie, and an emerging AI correcting its own operators — is the kind of evidence that a coach, mentor, or development programme could engage with in ways that generic feedback instruments cannot.

The facilitated debrief of a downloaded journal — used in a one-to-one coaching session, a cohort workshop, or a leadership development programme — is an obvious and promising next step. The engine produces the artefact. What happens with that artefact in structured facilitation contexts is an open question that the current beta has not yet answered.

### 8.5 The Open Architecture as Prospective Claim

Section 4.5 argues that the skin-modular architecture can be instantiated in any professional context where the relevant conditions hold. This claim is supported by three existing implementations across three fundamentally different domain taxonomies. It is not yet supported by a fourth.

The humanitarian response skin, the crisis negotiation skin, and the public health skin described in Section 4.5 do not exist. They are architectural projections — plausible instantiations of a modular design that has proven itself across three implementations. The projection may be correct. It has not been tested.

The authors flag this not to undermine the open architecture claim but to distinguish between what has been demonstrated and what is being proposed. The demonstration is three skins sharing an identical evaluation layer across entirely different domain layers. The proposal is that this generalises. The proposal is well-grounded. It remains a proposal.

### 8.6 What Would Constitute Failure

Academic honesty requires not only naming current limitations but specifying the conditions under which the paper's central claims would be falsified.

The central claim of this paper is that Last Prompt is a consequence-bearing reasoning environment that develops the cognitive capacity for consequential decision-making under partial information — a capacity that credentialism cannot confer, that bounce culture suppresses, that the KPI tunnel prevents from forming, and that the translation gap prevents from being expressed even when it exists.

This claim would be falsified if: practitioners who complete multiple runs of the engine show no measurable improvement in the rubric criteria that their chapter debriefs identify as weaknesses; if practitioners who score consistently Strong across an engine run perform no differently on real-world consequential decisions than practitioners who have never encountered the engine; if the AI evaluator's assessments show no meaningful correlation with independent human expert assessments of the same free-text plans; or if the environmental stat system proves insufficiently sensitive to distinguish between practitioners who are attending to the full environmental picture and those who are tunnel reasoning effectively within their domain.

These are the tests the system needs to pass. They have not yet been administered. The authors consider this the most important open question the paper raises, and the most important direction for future work.

### References

Abt, C. C. (1970). *Serious Games*. Viking Press.

Djaouti, D., Alvarez, J., & Jessel, J.-P. (2011). Classifying serious games: The G/P/S model. In P. Felicia (Ed.), *Handbook of Research on Improving Learning and Motivation through Educational Games: Multidisciplinary Approaches* (pp. 118–136). IGI Global.

Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3), 363–406.

Flavell, J. H. (1979). Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. *American Psychologist*, 34(10), 906–911.

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75–105.

Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.

Kozyrkov, C. (2019). Introduction to decision intelligence. *Towards Data Science*. https://medium.com/data-science/introduction-to-decision-intelligence-5d147ddab767

Lave, J., & Wenger, E. (1991). *Situated Learning: Legitimate Peripheral Participation*. Cambridge University Press.

Michael, D. R., & Chen, S. L. (2006). *Serious Games: Games That Educate, Train, and Inform*. Thomson Course Technology.

Mueller, B. (2024). *Reverse Engineering Reality: Observer Patch Holography*. FloatingPragma. Open access: https://oph-book.floatingpragma.io/

Mueller, B. (2024). Observer Patch Holography [GitHub repository]. FloatingPragma. https://github.com/FloatingPragma/observer-patch-holography

Schön, D. A. (1983). *The Reflective Practitioner: How Professionals Think in Action*. Basic Books.

Snowden, D. J., & Boone, M. E. (2007). A leader's framework for decision making. *Harvard Business Review*, 85(11), 68–76.

van Merriënboer, J. J. G., & Kirschner, P. A. (2018). *Ten Steps to Complex Learning: A Systematic Approach to Four-Component Instructional Design* (3rd ed.). Routledge.

*— End of Working Draft —*

*Last Prompt: Operationalising Partial Perspectives in Decision Intelligence Training*